

Immersive VR Training Simulation for Aircraft Maintenance

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Traditional training in aircraft maintenance struggles to meet the demands of modern aviation. Thus, in a collaboration with Malaysia Airlines Berhad Engineering Service, an immersive VR Training Simulation application was designed and developed to elevate the competencies of aircraft maintenance personnel.

PROBLEM STATEMENT:

- ⚠ Shortage of skilled aircraft maintenance personnel
- ⚠ Increasing complexity of aircraft systems and ensuring procedural accuracy.
- ⚠ Need for scalable and cost-effective training methods.
- ⚠ Maintenance training can be dangerous due to potential hazards like electrical shock.

SOLUTION AND NOVELTY:



Immersive VR application for training maintenance tasks.

- Realistic, interactive 3D scenarios in a virtual hangar environment.
- Hands-on practice without the need for physical aircraft or equipment.

BENEFIT & IMPACT:

- Improved safety through risk-free training
- Reduced training time and costs
- Enhanced learning retention and skills proficiency
- More flexible and scalable training capabilities

COLLABORATOR:



PRODUCT READINESS:

- Focuses on nose wheel assembly replacement procedure.
- Replicates real-world tools, environments, and scenarios with high fidelity.
- Enables trainees to master complex tasks in a safe, controlled setting.
- Allows repeated practice without wear on physical components.
- Scalable and consistent training across cohorts.
- Enhances aviation training by improving safety, precision, and performance.



TRL LEVEL:

▶ TRL 6



USEFULNESS:

- ▶ **Improved safety:** Identifies and addresses hazards early; enables safe practice of complex procedures.
- ▶ **Improved training efficiency:** Reduces training time and cost compared to traditional methods.
- ▶ **Increased knowledge retention:** Enhances retention through interactive and engaging learning.
- ▶ **Enhanced job satisfaction:** Enhanced job satisfaction: Provides a more enjoyable learning experience, boosting morale.